# Chapter 7 — Waves and Vibrations

## Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

**1.** Constructive interference path difference: (a) λ/8 (b) λ/4 (c) (2n+1)λ/2 (d) nλ

**2.** Distance between adjacent nodes: (a) λ (b) 2λ (c) λ/4 (d) λ/2

**3.** String fundamental frequency: (a) v/4ℓ (b) 2v/ℓ (c) v/2ℓ (d) v/ℓ

**4.** Closed pipe fundamental: (a) v/ℓ (b) 2v/ℓ (c) v/2ℓ (d) v/4ℓ

**5.** Closed pipe harmonics: (a) all (b) none (c) odd only (d) even only

**6.** Beat frequency equals: (a) $f_1 f_2$ (b) $|f_1 - f_2|$ (c) $(f_1+f_2)/2$ (d) $f_1 + f_2$

**7.** Source toward listener gives: (a) same frequency (b) zero frequency (c) higher frequency (d) lower frequency

**8.** Receding star shows: (a) green shift (b) red shift (c) blue shift (d) no shift

## Q.2 — SHORT QUESTIONS (6 × 2 = 12)

**1.** Why must sources be coherent to obtain a steady interference pattern?

**2.** Why is no energy transmitted past a node?

**3.** Why does a closed pipe sound lower than an open pipe of the same length?

**4.** Why do beats become hard to recognise above about 10 Hz?

**5.** Why does loading a tuning fork with wax lower its frequency?

**6.** How does the Doppler effect help astronomers measure stellar motion?

## Q.3 — LONG QUESTIONS (2 × 8 = 16)

**Q.3 (a)** Derive the frequencies of harmonics on a stretched string fixed at both ends. **(4)**

**(b)** Compare stationary waves in an open pipe and a pipe closed at one end. **(4)**

**Q.4 (a)** Explain the formation of beats and prove that beat frequency equals the difference of frequencies. **(4)**

**(b)** Explain the Doppler effect for a moving source and give its applications. **(4)**

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